

Project Proposal

Students: Md Shahanur Islam Shagor

Group No: ПИЦТ2-231-ОБ

Date: February 18, 2026

Topic: Decentralized Coordination and Acoustic Localization in Secure Autonomous Drone Swarms

1. Problem Statement

Most swarm systems degrade when a leader fails, when communication is unreliable, or when sensing is degraded by GPS-denied and vision-impaired conditions. This project proposes a secure autonomous drone swarm that maintains mission continuity using decentralized coordination, encrypted communication, and acoustic localization.

2. Project Motivation

Conventional architectures often rely on centralized control, weak telemetry integrity, and limited runtime fallback under latency or hardware issues. Real deployments in disaster response, inspection, and surveillance require continuity, trust, and fault tolerance even under partial failures.

3. Objectives

- Coordinate multiple drones through automatic leader election and follower synchronization.
- Preserve mission execution under leader loss, degraded motors, and network jitter.
- Protect swarm communication and event integrity using cryptographic mechanisms.
- Support operation in GPS-limited and vision-impaired environments through acoustic source localization.
- Provide simulation-first validation with optional real-drone integration.

4. Proposed System Overview

The system uses an event-driven swarm manager where one drone leads mission command flow while followers continuously exchange heartbeat, telemetry, and state updates. On leader failure, drones run a suitability-based election and promote a new leader without human intervention.

5. Core Technical Modules

- Swarm orchestration: role assignment, leader-follower command handling, mission state transitions.
- Dynamic obstacle avoidance: trajectory forecasting, collision-cone risk checks, and local rerouting.
- Acoustic localization: GCC-PHAT cross-correlation, TDOA estimation, least-squares source fusion.
- Secure communication: encrypted channels for inter-drone telemetry and control messages.
- Flying ledger: append-only signed event chain (SHA3-256 + Ed25519) for tamper-evident replication.
- Safety fallback: latency watchdog, local-avoidance fallback mode, emergency return/landing pathways.

6. Security and Reliability Strategy

1. Confidentiality and integrity are enforced through encrypted message exchange and authenticated event handling.
2. Distributed ledger replication reduces single-point trust assumptions across the swarm.
3. Runtime monitoring tracks RTT/jitter and triggers deterministic fallback behaviors when unsafe conditions are detected.
4. Motor-health checks and emergency logic prioritize safe mission continuation or controlled abort.

7. Hardware and Software Stack

Hardware: PX4-capable flight controller, companion computer (e.g., Raspberry Pi class), IMU/GPS, range sensors, radio module, audio sensors for acoustic sampling.

Software: Python swarm stack, PyQt GUI, PX4/MAVLink integration, optional C++ low-level control, ML modules for obstacle reasoning.

8. Methodology

5. Phase 1: Architecture design and simulation baseline.
6. Phase 2: Implement secure communication, leader election, and fault-handling logic.
7. Phase 3: Integrate dynamic obstacle avoidance and acoustic localization pipeline.
8. Phase 4: Add flying ledger verification and latency-aware fallback behavior.
9. Phase 5: Validate through test mode, scenario-based simulation, and optional real-drone trials.

9. Expected Outcomes

- A functional secure swarm prototype with decentralized coordination.
- Automatic leader replacement without mission restart.
- Acoustic source localization capability for degraded-visibility missions.
- Tamper-evident distributed telemetry/event records.
- Measurable resilience improvements under failure and latency stress scenarios.

10. Scope and Limitations

Scope: small-to-medium swarm experiments, simulation-first workflow, and controlled real-world testing.

Limitations: constrained onboard compute, non-military threat model, and dependence on sensor quality and communication conditions.

11. Conclusion

This proposal defines an integrated framework that combines decentralized coordination, security, acoustic intelligence, and safety fallback into one coherent swarm system. The work advances practical reliability beyond single-feature research prototypes.